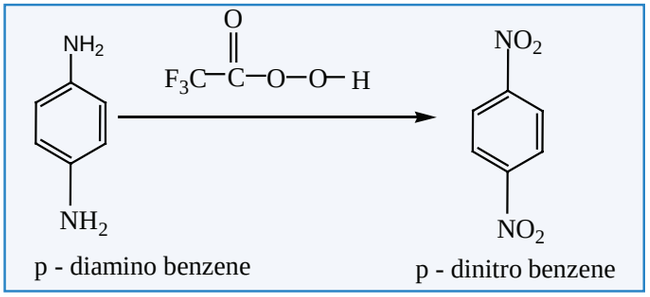
Amino group can be directly converted into nitro group, using caro's acid (H_2SO_5) (or) persulphuric acid ($H_2S_2O_8$) (or) peroxytrifluro acetic acid ($F_3C.CO_3H$) as oxidising agent.



13.1.7 Physical properties of nitro alkane

The lower nitroalkanes are colourless pleasant smelling liquids, sparingly soluble in water, but readily soluble in organic solvents like benzene, acetone etc… They have high boiling points because of their highly polar nature. Alkylnitrites have lower boiling points than nitro alkanes.

13.1.8 Chemical properties of nitroalkanes

Nitroalkanes undergo the following common reactions.

i. Reduction ii. Hydrolysis iii. Halogenations

i. Reduction of nitroalkanes

Reduction of nitroalkanes has important synthetic applications. The various reduction stages of nitro group are given below.

$$CH_3-NO_2 \xrightarrow[-H_2O]{+2[H]} CH_3-N=O \xrightarrow{+2[H]} CH_3-NHOH \xrightarrow[-H_2O]{+2[H]} CH_3-NH_2$$

nitromethane — nitroso methane — N - methyl hydroxylamine — methylamine

The final product depends upon the nature of reducing agent as well as the pH of the medium.

$$CH_3-NO_2 \xrightarrow[6[H]]{Sn/HCl} CH_3-NH_2 + 2H_2O \quad \text{(acid medium)}$$

Nitromethane — Methyl amine

$$CH_3-NO_2 \xrightarrow[4[H]]{Zn/NH_4Cl} CH_3-NHOH + H_2O \quad \text{(neutral medium)}$$

Nitromethane — N - methyl hydroxylamine

Reduction of alkyl nitrites

Ethylnitrite on reduction with Sn / HCl gives ethanol

$$CH_3CH_2-O-N=O + 6[H] \xrightarrow{Sn/HCl} CH_3CH_2-OH + NH_3 + H_2O$$

ii. Hydrolysis of nitroalkanes

Hydrolysis can be effected using conc. HCl or conc. H_2SO_4. Primary nitroalkanes on hydrolysis gives carboxylic acid, and the secondary nitroalkanes give ketones. The tertiary nitroalkanes have no reaction.

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